HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Supported airport master planning and terminal layout analysis, developing **3D visualisations** of proposed infrastructure changes for client presentations.
- Developed and maintained **analytical models** in Excel and Python to forecast passenger flow and assess capacity constraints for terminal operations, impacting planning for up to **50 million passengers** annually.
- Contributed to technical reports and business cases for airport expansion projects by gathering and analysing operational data, including **flight schedules** and passenger throughput.
- Drove innovation by integrating emerging **AI technologies** into operational forecasting models, enhancing predictive accuracy for passenger demand by an estimated **15%**.
- Automated data ingestion and reporting processes for airport operational metrics using **Python scripting**, reducing manual effort by approximately **20 hours per week** for the advisory team.
- Engaged with international clients and stakeholders through workshops and technical reviews to define requirements for **digital transformation** initiatives within airport operations.
- Containerised analytical applications using **Docker** and deployed them to a cloud environment, ensuring consistent execution across development and client-facing deployments.

SKILLS

Analytics & Modelling – Python, Excel, Power BI, SQL, Data Warehousing, Statistical Analysis, Operational Research, Capacity Analysis, Forecasting, Predictive Analytics, AI Technologies

Airport Planning – Master Planning, Terminal Planning, Airfield Planning, Operational Readiness, Passenger Flow Modelling, Capacity-Demand Assessment, Long-Term Expansion Strategies

Programming Languages – Python, SQL, R

Tools & Platforms – Power BI, Tableau, Docker, AWS (EC2, S3, Lambda, CloudWatch), Jenkins, Git, Jira, Confluence

Data Analysis & BI – Data Modelling, DAX, Power Query, Data Visualisation, Reporting, Business Intelligence

PROJECTS

Airport Capacity Analysis Platform

- Developed a Python-based application to model airport terminal capacity, processing historical **flight data** and passenger load factors to generate utilisation forecasts.
- Integrated **Power BI** for interactive dashboards visualising key capacity metrics and identifying potential bottlenecks across different operational scenarios, supporting **12** distinct planning modules.
- Built REST APIs using **Flask** to serve capacity data to downstream planning tools, enabling real-time scenario analysis for airport infrastructure development.

Automated Airport Operations Reporting

- Created automated data pipelines using **Python** and **SQL** to extract and transform operational data from disparate sources, consolidating metrics for daily reporting.
- Designed and implemented a suite of **Power BI** reports to present key performance indicators related to passenger flow, gate utilisation, and baggage handling efficiency to operations teams.
- Managed the data warehousing aspects of the reporting solution, ensuring data integrity and optimising **SQL queries** for timely report generation, reducing refresh times by **40%**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)